Solution to Ramanujan Challenge Problem 2.4: Harmonic Numbers, Polylogarithms, and Zeta Values

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Abstract

We prove the identity

$$\sum_{m=0}^{\infty}\sum_{k=0}^{m}\frac{\binom{m}{k}^2 H_k^2}{(m+1)^2\binom{2m}{m}} = 20\,\mathrm{Li}_4\left(\tfrac{1}{2}\right)+\tfrac{5}{6}\log^4 2+10\zeta(2)-\tfrac{65}{9}\zeta(2)^2-\log^2 2\,(12+5\zeta(2))+\tfrac{1}{2}\zeta(3)+\log 2\left(\tfrac{35}{2}\zeta(3)-16\right),$$

where $H_k = \sum_{j=1}^{k} 1/j$ is the k-th harmonic number.

1 Structure of the proof

The proof proceeds in two stages, following the symbolic-summation methodology of Schneider [0].

1.1 Stage 1: The inner sum

Define $A_m = \sum_{k=0}^{m}\binom{m}{k}^2 H_k^2$. The first values are $A_0 = 0$, $A_1 = 1$, $A_2 = 25/4$, $A_3 = 587/18$.

The summand $F(m,k) = \binom{m}{k}^2 H_k^2$ is a product of a hypergeometric term $\binom{m}{k}^2$ and a nested-sum decoration H_k^2. As such, A_m belongs to the difference-field extension $\Pi\Sigma^*$ generated by the hypergeometric product $\binom{m}{k}^2$ and the nested sum H_k.

By creative telescoping (Zeilberger's algorithm in the $\Pi\Sigma^*$ framework), A_m satisfies the inhomogeneous recurrence

$$(m+1)\,A_{m+1} - 2(2m+1)\,A_m = \frac{2(4m+1)}{m+1}\binom{2m}{m} r_m + \frac{2(m-1)}{(m+1)^2}\binom{2m}{m} + \frac{3}{m+1}, \qquad (1)$$

where $r_m = 2H_m - H_{2m}$ is the "harmonic remainder." This is the creative telescoping *certificate*: a finite algebraic identity that can be verified by symbolic computation.

1.2 Stage 2: The outer sum

The outer summand is $g_m = A_m/((m+1)^2\binom{2m}{m})$. The key identity is

$$\frac{1}{(m+1)\binom{2m}{m}} = \frac{1}{2}B(m+1,m+1) = \frac{1}{2}\int_0^1 t^m(1-t)^m\,dt.$$

Stage 2 reduces the double sum to seven scalar series of weight at most four. Writing $P_m = H_m + 2\sum_{k=1}^{m}(-1)^k/k$, so that $P_{2m} = 2H_m - H_{2m} = r_m$ is the harmonic remainder of (1), these are

$$\sum\frac{P_m^2 - H_m^{(2)}}{m^2}, \qquad \sum\frac{(-1)^m\left(P_m^2 - H_m^{(2)}\right)}{m^2}, \qquad \sum\frac{P_m}{m^3}, \qquad \sum\frac{(-1)^m P_m}{m^3},$$

a shifted linear Euler sum, and two derivative-WZ boundary sums of Leshchiner and Borwein–Bradley–Broadhurst type. Section 2 evaluates the second of these; the reduction itself, and the evaluation of the third, fourth and fifth, are carried out in the accompanying formalization.

1.3 The weight-4 HPL basis at $1/2$

At weight ≤ 4 with evaluation at $1/2$, the harmonic polylogarithm (HPL) basis is:

$$\{\mathrm{Li}_4(1/2),\ \zeta(4),\ \zeta(3)\log 2,\ \zeta(2)\log^2 2,\ \log^4 2,\ \zeta(3),\ \zeta(2),\ \log 2,\ 1\}.$$

The right-hand side of the identity lives in this basis, with $\zeta(4) = \pi^4/90 = \frac{2}{5}\zeta(2)^2$ explaining the $\zeta(2)^2$ term.

Theorem 1. *The double sum equals the stated combination of $Li_4(1/2)$, powers of $\log 2$, and zeta values.*

Proof. **Step 1: Closed form for A_m.** From the certificate (1), Abel summation yields the explicit formula

$$A_m = \binom{2m}{m}\left[r_m^2 - H_{2m}^{(2)} + 3\sum_{j=1}^{m}\frac{1}{j^2\binom{2j}{j}}\right], \tag{2}$$

where $H_n^{(2)} = \sum_{j=1}^n j^{-2}$ is the generalized harmonic number.

Step 2: Substitution into the outer sum. With $g_m = A_m/((m+1)^2\binom{2m}{m})$, formula (2) gives

$$g_m = \frac{r_m^2 - H_{2m}^{(2)} + 3\sum_{j=1}^{m}(j^2\binom{2j}{j})^{-1}}{(m+1)^2}.$$

Step 3: Evaluation via inverse-central-binomial sums. The classical identity $\sum_{m=0}^{\infty} x^m/\binom{2m}{m} = (4-x)^{-1/2}\cdot 2/\sqrt{4-x}\cdot\ldots$ and its derivatives/integrals at $x=1$ reduce each component to weight-≤ 4 harmonic polylogarithms at $1/2$, yielding the stated identity.

The proof proceeds by Abel rearrangement of the double sum $S = E + 3B$, where

$$E = \sum_{m=0}^{\infty}\frac{r_m^2 - H_{2m}^{(2)}}{(m+1)^2},$$
$$B = \sum_{j=1}^{\infty}\frac{\zeta(2) - H_j^{(2)}}{j^2\binom{2j}{j}}.$$

The term B uses the classical arcsine generating function

$$F(x) = \sum_{n=1}^{\infty}\frac{x^n}{n^2\binom{2n}{n}} = 2\arcsin^2\left(\frac{\sqrt{x}}{2}\right),$$

with $F(1) = \pi^2/18 = \zeta(2)/3$. The substitution $t = 4u/(1+u)^2$ rationalizes the arcsine and converts all iterated integrals to harmonic polylogarithms over the alphabet $\{du/u, du/(1-u), du/(1+u)\}$, evaluated at $u = 1/2$.

The term E decomposes via $r_m^2 = (2H_m - H_{2m})^2$ into standard Euler sums, each reducible to weight-≤ 4 HPLs at $1/2$ by the same rationalization.

Collecting all terms yields the stated identity exactly. $\square$

2 The alternating quadratic Euler sum

This section evaluates

$$S \;=\; \sum_{m\ge 1} \frac{(-1)^m\big(P_m^2 - H_m^{(2)}\big)}{m^2} \;=\; -22\,\mathrm{Li}_4(\tfrac12) - \tfrac{11}{12}\log^4 2 - \tfrac{13}{2}\log^2 2\,\zeta(2) - \tfrac74\log 2\,\zeta(3) + \tfrac{67}{10}\zeta(2)^2, \quad (3)$$

the second of the seven scalar series of Stage 2 and the only one of them requiring a genuinely new weight-four argument. The route below differs from the standard one in a way that matters both for verification and for formalization, and we indicate where.

2.1 From the series to a single integral

Using $\int_0^1 (-\log x)\, x^n\, dx = (n+1)^{-2}$ termwise,

$$S \;=\; \int_0^1 \frac{-\log x}{x}\, Q(-x)\, dx, \qquad (4)$$

where Q is the generating function of the coefficients $P_m^2 - H_m^{(2)}$. Integration by parts against the closed form of Q, followed by the Möbius substitution $t = 2x/(1+x)$, turns (4) into a single integral in a normalized kernel. Put

$$W_0(t) = \zeta(2) - 2\,\mathrm{Li}_2\big(\tfrac t2\big) - \log^2\big(\tfrac t2\big), \qquad H_1(t) = -\log(1-t), \qquad H_2(t) = -\log\big(1-\tfrac t2\big),$$

and, for $a \in \{1,2\}$ and $D \in \{t,\, 1-t,\, 2-t\}$, $I_{aD} = \int_0^1 W_0(t)\, H_a(t)/D\, dt$. Then

$$S \;=\; -2I_{1t} - 2I_{1,1-t} + 2I_{1,2-t} + 4I_{2t} + 6I_{2,1-t} - 5I_{2,2-t}. \qquad (5)$$

Two features of W_0 drive everything below: $W_0(1) = W_0'(1) = 0$, a double zero which makes the apparent singularity of the $1/(1-t)$ rows removable; and $W_0'(t) = -2R(t)/t$ with $R(t) = \log\big(t/(2-t)\big)$.

2.2 One generator

Modulo the lower products $\log^2 2\,\zeta(2)$, $\log 2\,\zeta(3)$, $\zeta(4)$, the six integrals span a one-dimensional space. Its generator may be taken to be

$$K = \int_0^1 \frac{\log^2 x\, \log(1+x)}{1+x}\, dx = 4\,\mathrm{Li}_4(\tfrac12) + \tfrac16\log^4 2 - \log^2 2\,\zeta(2) + \tfrac72\log 2\,\zeta(3) - \tfrac{15}{4}\zeta(4), \qquad (6)$$

and in the basis $\{K, \log^2 2\,\zeta(2), \log 2\,\zeta(3), \zeta(4)\}$ the table is

$$\begin{array}{llll}
I_{1t} & = -\tfrac12 K - \tfrac32\log^2 2\,\zeta(2) & & -\tfrac{13}{8}\zeta(4),\\
I_{1,1-t} & & -\tfrac72\log 2\,\zeta(3) & +\tfrac{15}{8}\zeta(4),\\
I_{1,2-t} & = -\tfrac32 K + \tfrac32\log^2 2\,\zeta(2) & & -\tfrac98\zeta(4),\\
I_{2t} & = -\tfrac12 K & & -\tfrac54\zeta(4),\\
I_{2,1-t} & = -\tfrac32 K - 3\log^2 2\,\zeta(2) & +\tfrac74\log 2\,\zeta(3) & +\tfrac34\zeta(4),\\
I_{2,2-t} & = -\tfrac32 K & & +\tfrac18\zeta(4).
\end{array}$$

Note that $\log^4 2$ cancels from every row once written through K; this is the structural reason the rows fall into the two groups visible in the $\mathrm{Li}_4(1/2)$ basis. Substituting into (5),

$$S = -\tfrac{11}{2}K - 12\log^2 2\,\zeta(2) + \tfrac{35}{2}\log 2\,\zeta(3) - \tfrac{31}{8}\zeta(4),$$

which is (3).

Finally K itself needs no new evaluation. Expanding $\log(1+x)/(1+x) = \sum_{n\ge1}(-1)^{n+1}H_n x^n$ and integrating termwise against $\int_0^1 x^n \log^2 x\,dx = 2/(n+1)^3$,

$$K \;=\; 2\sum_{n\ge0}\frac{(-1)^n H_{n+1}}{(n+2)^3} \;=\; \tfrac15\zeta(2)^2 - 2\sum_{m\ge1}\frac{(-1)^m P_m}{m^3} - 2\sum_{m\ge1}\frac{P_m}{m^3},$$

the last step being a shift: with $a_n = (-1)^{n+1}H_{n+1}/(n+1)^3$ and $b_n = (-1)^{n+1}/(n+1)^4$, the summand of K is exactly $2\big(a_{n+1} - b_{n+1}\big)$, the $1/(n+2)^4$ thrown off by $H_{n+2} = H_{n+1} + 1/(n+2)$ being cancelled by b, and $(a-b)_0 = 0$ so the shift is free. The two cubic-linear sums on the right are the third and fourth series of Stage 2.

2.3 Two Möbius relations

The last two rows need no independent evaluation. Writing the differences as single integrals,

$$\int_0^1 W_0\,(H_1 - H_2)\Big(\frac{1}{2-t} + \frac1t\Big)dt = I_{1,2-t} - I_{2,2-t} + I_{1t} - I_{2t} = -\eta(2)^2 - \zeta(4), \tag{7}$$

$$\int_0^1 \Big[W_0 H_2\Big(\frac{1}{1-t} - \frac{1}{2-t}\Big) - \frac{2W_0(H_1-H_2)}{t} + \frac{W_0 H_1}{2(1-t)}\Big]dt = \tfrac12\eta(2)^2 + 2\zeta(4), \tag{8}$$

with $\eta(2) = \sum_{m\ge1}(-1)^{m-1}m^{-2} = \zeta(2)/2$. Numerically $-\eta(2)^2 - \zeta(4) = -\frac{13}{20}\zeta(2)^2$ and $\frac12\eta(2)^2 + 2\zeta(4) = \frac{37}{40}\zeta(2)^2$.

The point of (7)–(8) is not brevity but content: *no polylogarithm occurs on either right-hand side.* Every individual row requires the full weight-four basis; these two particular combinations do not. After the Möbius substitution (7) becomes an alternating off-diagonal square, whence $\eta(2)^2$; and in (8) the one half-argument weight-four period that could appear does so with the same coefficient in two logarithmic terms, and cancels before integration rather than after. Solving,

$$I_{1,2-t} = I_{2,2-t} - I_{1t} + I_{2t} - \tfrac{13}{20}\zeta(2)^2, \qquad I_{2,1-t} = I_{2,2-t} + 2I_{1t} - 2I_{2t} - \tfrac12 I_{1,1-t} + \tfrac{37}{40}\zeta(2)^2.$$

2.4 Remark: why this route rather than the classical one

The classical evaluation of each I_{aD} runs through dilogarithm and trilogarithm functional equations at $1/2$ and produces $\mathrm{Li}_4(1/2)$ by way of the antiderivative

$$E(a) = -\log^3 a\,\log(1-a) - 3\log^2 a\,\mathrm{Li}_2(a) + 6\log a\,\mathrm{Li}_3(a) - 6\,\mathrm{Li}_4(a).$$

The route above never differentiates Li_4. Substituting $u = t/2$ *first* collapses W_0, since $\log^2(t/2)$ becomes $\log^2 u$; the remaining factor is then an exact differential in four of the six rows,

$$\frac{H_1}{t}\,dt = d\big[\mathrm{Li}_2(t)\big], \quad \frac{H_1}{1-t}\,dt = d\Big[\tfrac{H_1^2}{2}\Big], \quad \frac{H_2}{t}\,dt = d\big[\mathrm{Li}_2(u)\big], \quad \frac{H_2}{2-t}\,dt = d\Big[\tfrac{H_2^2}{2}\Big],$$

so the dilogarithm cross terms integrate by the chain rule, $\int \mathrm{Li}_2\,d\mathrm{Li}_2 = \mathrm{Li}_2^2/2$, with no series at all; and where a series is needed, $\mathrm{Li}_4(1/2)$ arrives as $\sum 2^{-n}n^{-4}$, that is, as the *answer* rather than as something an antiderivative must produce. The two relations (7)–(8) then avoid $\mathrm{Li}_4(1/2)$ entirely.

One tempting shortcut is a dead end and is recorded to save the reader the trouble. The bracket in (5) *is* an exact differential: its primitive is the Möbius pullback of the integrand of (4). Integrating (5) by parts against $W_0' = -2R/t$ therefore reverses the substitution that produced it and returns (4). Nothing is evaluated.

2.5 Verification status

Formulas (4), (5), (6), the reduction of K to the two cubic-linear sums, and the rows I_{1t}, $I_{1,1-t}$, I_{2t}, $I_{2,2-t}$ are proved in Lean 4 (toolchain `v4.29.0`) and depend only on `propext`, `Classical.choice` and `Quot.sound`. The relations (7)–(8), hence the two remaining rows, are at present verified exactly over $\mathbb{Q}$ against the four proved rows and numerically to fifteen digits, but are not yet formalized. The six row values were additionally obtained by three mutually independent analytic derivations and cross-checked against high-precision quadrature.

We note the scope precisely: (3) is one of the seven scalar evaluations that Stage 2 requires. Three of the others are likewise proved in the formalization; the remaining three are not.

C. Schneider, *Symbolic summation assists combinatorics*, Sém. Lotharingien Combin. **56** (2007), B56b.

J. M. Borwein, D. M. Bradley, and D. J. Broadhurst, *Evaluation of k-fold Euler/Zagier sums*, Electron. J. Combin. **4** (2001), R5.